

Mechanics Extras — Friction (Extra Material)

Unit 3 — Laws of Motion

These notes add worked examples on friction to Unit 3 of the Physics syllabus. Friction is the resistive force that acts parallel to a surface, opposing the motion or attempted motion of an object. Kinetic friction acts while an object slides.

Worked Example: Box on a Floor

A 20 kg box is pushed across a level floor with a horizontal force of 60 N. The friction between box and floor is 40 N. The net force is $60 - 40 = 20$ N, so by Newton's second law the acceleration is $a = F_{\text{net}} / m = 20 / 20 = 1 \text{ m/s}^2$.

When Does Friction Reverse?

If the pushed force is removed, friction becomes the only horizontal force and brings the box to rest. Doubling the mass doubles the normal reaction and therefore roughly doubles the friction (friction is proportional to the normal force for sliding objects).

Exam-Style Question

A 5 kg crate rests on a ramp. Its weight is $W = mg = 5 \times 9.8 = 49$ N. State which force balances the component of weight acting down the ramp, and explain what happens to that force as the ramp is tilted more steeply.